

Code No: 134AP

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**B. Tech II Year II Semester Examinations, April/May - 2023****DATABASE MANAGEMENT SYSTEMS**

(Common to CSE, IT)

Time: 3 Hours

Max. Marks: 75

Note: i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) What is a relation? Differentiate between a relation schema and a relation instance. [2]
- b) What is the difference between a candidate key and the primary key for a given relation? What is a super key? [3]
- c) What is Trigger? Why it is used? [2]
- d) Describe the projection operator. What can you say about the cardinality of the input and output tables for this operator? [3]
- e) What are the problems related to decompositions? [2]
- f) Define 'Functional Dependencies'. How are primary keys related to FDs? [3]
- g) What functionality does the recovery manager of a DBMS provide? [2]
- h) Define the terms transaction, schedule, complete schedule, and serial schedule. [3]
- i) What is the search key for an index? [2]
- j) What is an index? What is the relationship between file and indexes? Can we have several indexes on a single file of records? [3]

PART – B**(50 Marks)**

- 2.a) Explain i) The data definition language ii) The data manipulation language iii) The buffer manager iv) The data model.
- b) What are integrity constraints? Define the terms primary key constraint and foreign key constraint. How are these constraints expressed in SQL? What other kinds of constraints can we express in SQL? [5+5]

OR

- 3.a) What guidelines would you use for each of these choices when doing ER design. Whether to use an attribute or an entity set, an entity or a relationship set, a binary or ternary relationship, or aggregation.
- b) A company database needs to store information about employees (identified by ssn, with salary and phone as attributes), departments (identified by dn, with dname and budget as attributes), and children of employees (with name and age as attributes). Employees work in departments; each department is managed by an employee; a child must be identified uniquely by name when the parent (who is an employee; assume that only one parent works for the company) is known. We are not interested in information about a child once the parent leaves the company. Draw an ER diagram that captures this information. [5+5]

4.a) Describe the set operations of relational algebra, including union($\cup$), intersection($\cap$), set-difference($-$), and cross-product($\times$). For each, what can you say about the cardinality of their input and output tables?

b) What are nested queries? What is correlation in nested queries? How would you use the operators IN, EXISTS, UNIQUE, ANY, and ALL in writing nested queries? Why are they useful? [4+6]

OR

5.a) What aggregate operators does SQL support?

b) Consider the following relational schema. An employee can work in more than one department, the pct_time field of the Works relation shows the percentage of time that a given employee works in a given department.

Emp(eid: integer, ename: string, age: integer, salary: real)

Works(eid: integer, did: integer, pct_time: integer)

Dept(did: integer, budget: real, managerid: integer)

Write SQL integrity constraints (domain, key, foreign key, or CHECK constraints; or assertions) or SQL triggers to ensure each of the following requirements, considered independently.

i) Employees must make a minimum salary of \$1000.

ii) Every manager must be also be an employee.

iii) A manager must always have a higher salary than any employee that he or she manages.

iv) Whenever an employee is given a raise, the department's budget must be increased to be greater than the sum of salaries of all employees in the department. [4+6]

6.a) What are normal forms? What is their purpose?

b) Define 1NF and 2NF. What is the motivation for putting a relation in BCNF? What is the motivation for 3NF? [3+7]

OR

7. Define multi-valued dependencies, explain 4NF. [10]

8.a) Explain how the use of Strict 2PL would prevent interference between the two transactions.

b) What is the phantom problem? What impact does it have on performance? [4+6]

OR

9. Discuss about remote backup systems. [10]

10. Discuss in detail about ISAM. [10]

OR

11. Explain extendible hashing with example. [10]

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R16**Code No: 134BU****JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech II Year II Semester Examinations, April/May - 2023****OPERATING SYSTEMS****(Common to CSE, IT)****Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) Draw the structure of OS. [2]
- b) Write the objectives of operating system. [3]
- c) Differentiate between process and thread. [2]
- d) What is context switch? [3]
- e) How the segmentation differ from paging? [2]
- f) What is Virtual memory? [3]
- g) What is File sharing? [2]
- h) Describe the importance of seek(). [3]
- i) Which are the necessary conditions for Deadlock? [2]
- j) What is language based protection? [3]

PART – B**(50 Marks)**

- 2.a) Explain about the operations of operating systems.
 - b) Explain how operating system used in a variety of computing environments. [5+5]
- OR**
- 3.a) Write the services provided by Operating System in terms of security and protection.
 - b) What is operating system? Explain the abstract view of the components of a computer system. [5+5]
- 4.a) What is monitor and give a solution for Reader's and writer's Critical section problem using monitors.
 - b) Explain about the process states with neat sketch. [5+5]
- OR**
- 5.a) What is Scheduler? Explain the queuing diagram representation of process scheduler with neat sketch.
 - b) Explain about the process control block with neat sketch. [5+5]

6. For the following page reference string 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1. Calculate the page faults applying the following Page replacement algorithms for a memory with three frames:

- a) LRU
- b) FIFO.

[5+5]

OR

- 7.a) What is Demand paging? Explain the working of demand paging and also give the advantages and disadvantages of Demand Paging.
- b) Explain the causes of Thrashing.

[7+3]

- 8.a) Explain about FCFS disk scheduling algorithms with an example.
- b) Explain about the file system structure.

[5+5]

OR

- 9.a) Explain about the various file access methods.
- b) Explain about free space management in file system.

[5+5]

- 10.a) Explain the protection mechanism illustrating the use of protection domain and access control list.
- b) Explain the implementation of the Access Matrix.

[5+5]

OR

11. Explain the methods for handling deadlocks.

[10]

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Code No: 134AK

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**B. Tech II Year II Semester Examinations, April/May-2023****COMPUTER ORGANIZATION**

(Common to CSE, IT)

Time: 3 Hours

Max. Marks: 75

Note: i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

- 1.a) Define computer architecture. [2]
- b) Why is CPU known as the heart of the computer? [3]
- c) What are assembly directives? [2]
- d) List the main advantages of segmented memory scheme. [3]
- e) What is a macro? [2]
- f) What are the different DEBUG command characters used in MASM? [3]
- g) What is daisy chaining interrupt? [2]
- h) What are the three ways of representing a negative fixed-point binary numbers? [3]
- i) What is meant by auxiliary memory? [2]
- j) Give examples for parallel processing. [3]

PART – B

(50 Marks)

2. With a suitable flowchart describe the concept of instruction cycle. [10]
- OR**
3. Demonstrate the general configuration of a micro programmed control unit. [10]
4. Describe general bus operation cycle of 8086 with suitable timing diagram. [10]
- OR**
5. Illustrate the functionality of the following instructions of 8086:
(a) AAM (b) LDS (c) ADC (d) XSLT (e) LODS. [10]
6. How to write an assembly language program? Explain each step in detail with a suitable program. [10]
- OR**
7. Write a program to find out the number of even numbers and odd numbers from a given series of 16-bit hexadecimal numbers. [10]
8. With the help of a flowchart explain the division operation on floating point numbers. [10]
- OR**
- 9.a) Make a comparison of isolated I/O with memory-mapped I/O. [5+5]
- b) Describe an I/O interface unit.

10. Explain the three mapping procedures used in cache memory organization with suitable examples. [10]

OR

11. Demonstrate the usage of crossbar switch in multiprocessor systems with necessary block diagram. [10]

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UNIVERSITY Used papers 2023

Code No: 135BM

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**B. Tech III Year I Semester Examinations, January/February - 2023****SOFTWARE ENGINEERING**

(Common to CSE, IT)

Time: 3 Hours

Max. Marks: 75

Note: i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

- 1.a) What is legacy software? [2]
- b) Explain about the waterfall model. [3]
- c) What is function requirement? [2]
- d) Describe data models. [3]
- e) What are architectural styles? [2]
- f) Explain about object constraint language. [3]
- g) Define integration testing. [2]
- h) Explain about metrics for source code. [3]
- i) What is SQA? [2]
- j) Discuss about RMMM plan. [3]

PART – B

(50 Marks)

2. Define process model. What are evolutionary process models? Explain in detail. [10]
OR
- 3.a) What are software myths? Explain in detail.
b) Define software engineering. Explain about process patterns. [5+5]
4. Explain in detail about requirements elicitation and analysis. [10]
OR
- 5.a) What are feasibility studies? Explain.
b) Discuss in detail about structured methods. [5+5]
- 6.a) Explain about task analysis and modeling.
b) Discuss about program design language. [5+5]
OR
- 7.a) What is architecture? Why is architecture important?
b) Describe architectural design and interface design elements. [5+5]
8. Explain in detail about white-box and black-box testing techniques. [10]
OR
- 9.a) What is software measurement? Explain in detail.
b) Discuss about metrics for analysis model. [5+5]

10. Explain the following:
a) Risk identification
b) ISO 9000 quality standards.

[5+5]

OR

11. Explain the following:
a) Six Sigma quality standard
b) Software risks.

[5+5]

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UNIUH Used papers 2023

Code No: 152AF**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B.Tech I Year II Semester Examinations, June - 2022****PROGRAMMING FOR PROBLEM SOLVING****(Common to EEE, CSE, IT, CSIT, ITE, CE(SE), CSE(CS), CSE(DS), CSE(Networks))****Time: 3 Hours****Max. Marks: 75**

Answer any five questions
All questions carry equal marks

...

- 1.a) Explain briefly about the components of a computer.
b) Write an algorithm to find the factorial of a given number. [8+7]
2. Define operator? List and explain various operators used in C programming language. [15]
- 3.a) Develop a C Program to perform matrix multiplication.
b) What is the syntax for Structure? Explain with an example. [7+8]
- 4.a) Explain in detail about various types of string handling functions with an example.
b) Describe briefly about Enumerated Data Types. [8+7]
- 5.a) Explain briefly about Commonly used Preprocessor commands include and define.
b) Develop a C program to write contents in to file using file I/O. [8+7]
- 6.a) Illustrate recursive function with a suitable example.
b) Differentiate call by value and call by reference. [7+8]
- 7.a) Explain in detail about user defined functions with an example program.
b) Develop a C program to print the Fibonacci series using recursion. [8+7]
- 8.a) Write an algorithm to find the roots of a quadratic equations.
b) Develop a C program to search an element using Binary search technique. [7+8]

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Code No: 152AG

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**B.Tech I Year II Semester Examinations, June - 2022****ENGINEERING GRAPHICS****(Computer Science and Engineering)****Time: 3 Hours****Max. Marks: 75****Answer any three questions
All questions carry equal marks**

...

- 1.a) Draw a parabola if the distance of the focus from the directrix is 60 mm.
b) A circle of diameter 40 mm rolls inside another circle of radius 60 mm. Draw the hypocycloid traced by a point on the rolling circle initially in contact with the directing circle for one revolution. [10+15]
- 2.a) A line AB, 50 mm long, has its end A at 40 mm above the HP and 20 mm in front of the VP. The end B is close to the HP but away from the VP. Draw the projections of the line if it is inclined to the HP at 30° and to the VP at 45° .
b) A rectangle ABCD of size 30 mm x 20 mm is inclined to the HP at 30° . Its shorter side AB is parallel to the HP and inclined at 45° to the VP. Draw the projections of the rectangle. [10+15]
- 3.a) A triangular prism, 40 mm side of base and 60 mm length of axis, has its axis perpendicular to the VP. Draw the projections if one of the rectangular faces is parallel to the HP and 20 mm above the HP.
b) A cone of base 60 mm diameter and height 80 mm is resting on a point on the circumference of base on the HP with its apex 55 mm above the HP. Draw its projections if its axis is inclined at 45° to the VP. [10+15]
4. A vertical square prism with a 60 mm base side and an 80 mm axis length, is completely penetrated by a horizontal square prism with a 40 mm side base and a 100 mm axis length such that their axes bisect each other. The faces of both the prisms are equally inclined to the VP. Draw the three views of the solids showing LOI. [25]

5. Draw (a) Front view (b) Top view and (c) side view of the isometric view given in the figure 1 (All dimensions are in mm). [25]

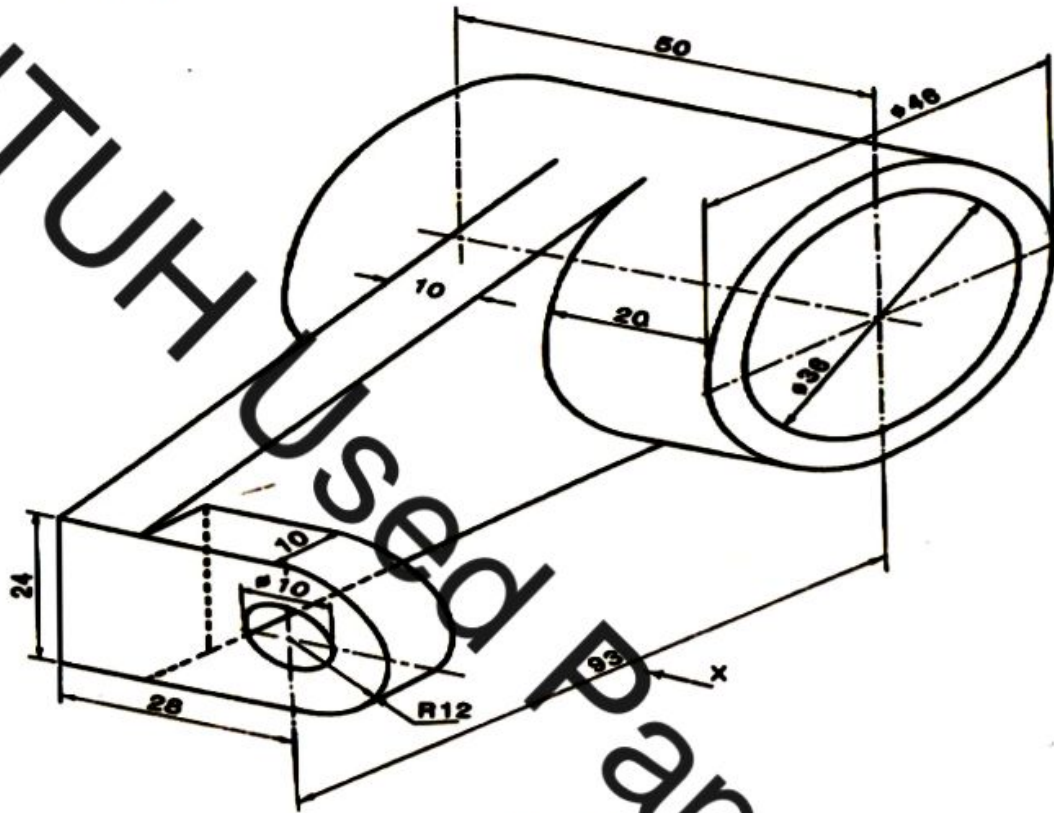


Figure 1

6. Figure 2 below shows the FV and TV of the object. Draw its isometric view. (All dimensions are in mm). [25]

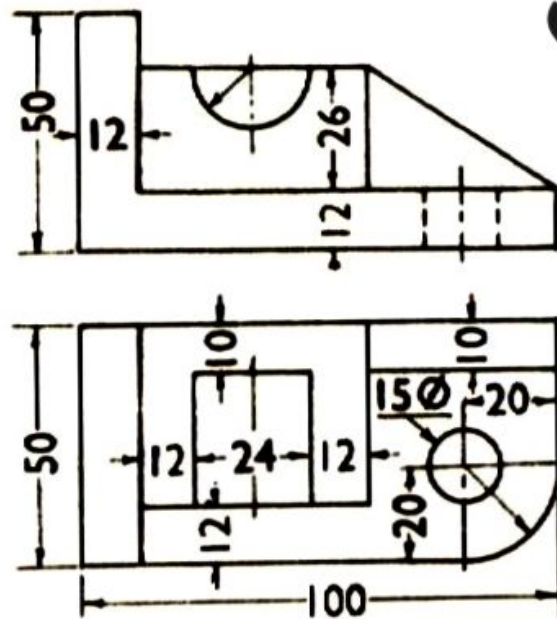


Figure 2

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Code No: 152AE

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**B.Tech I Year II Semester Examinations, June - 2022****APPLIED PHYSICS****(Common to EEE, CSE, IT, CSIT, ITE, CE(SE), CSE(CS), CSE(DS), CSE(Networks))****Time: 3 Hours****Max. Marks: 75**

Answer any five questions
All questions carry equal marks

...

- 1.a) Derive an expression for the wavelength λ of the matter waves.
b) Describe an experiment to verify the existence of matter waves. [5+10]
- 2.a) Explain Heisenberg's Uncertainty principle of position and momentum variables.
b) Write a note on wave particle duality and properties of matter waves. [8+7]
- 3.a) What are intrinsic and extrinsic semiconductors?
b) Derive an expression for Fermi level in an intrinsic semiconductor and hence show that the Fermi level lies exactly in the middle of the forbidden band. [5+10]
- 4.a) What is a photo detector? Explain the working principle of semiconductor photodiode.
b) When 3×10^{11} photons each with wavelength of $0.85\mu\text{m}$ are incident on a photodiode, on average 1.2×10^{11} electrons are generated. Determine the quantum efficiency and responsivity. [8+7]
- 5.a) Explain the construction, principle and working of DFB laser.
b) Explain about the different modes that are propagated through step-index and graded index fiber? [10+5]
- 6.a) Discuss the concept of Acceptance angle and Acceptance cone of a fiber.
b) Derive a relation between acceptance angle and the refractive indices of core and cladding materials. [8+7]
- 7.a) What is dielectric polarization? Describe briefly types of polarizations.
b) Derive Clausius-Mosotti relation for dielectric material. [6+9]
- 8.a) Explain the classification of magnetic materials.
b) When a magnetic material is subjected to a magnetic field of intensity 250 A/m , its relative permeability is 15. Calculate its magnetization and magnetic flux density. Given that $\mu_0 = 4\pi \times 10^{-7}$. [9+6]

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Code No: 151AG

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**B. Tech I Year I Semester Examinations, March/April - 2023****BASIC ELECTRICAL ENGINEERING****(Common to EEE, CSE, IT, CSIT, ITE, CE(SE), CSE(CS), CSE(DS), CSE(N), CSD)****Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A**(25 Marks)**

- 1.a) State Norton's Theorem. [2]
- b) Write short notes on voltage source and current source. [3]
- c) Define power factor. What is the ideal value of power factor? [2]
- d) What is balanced system and unbalanced system? [3]
- e) Write the applications of an auto transformer. [2]
- f) What is meant by ideal transformer? What are the properties of ideal transformer? [3]
- g) Write the applications of single-phase induction motor? [2]
- h) A 3 phase 4 pole, 50 Hz induction motor is running at 1455 rpm. Find the slip speed and slip. [3]
- i) What is the component used in L.T switch gear. [2]
- j) Define earthing, explain its importance? [3]

PART - B**(50 Marks)**

- 2.a) Using Thevenin's theorem, calculate Thevenin's Resistance, Thevenin's voltage across 5 ohms resistor (figure 1).

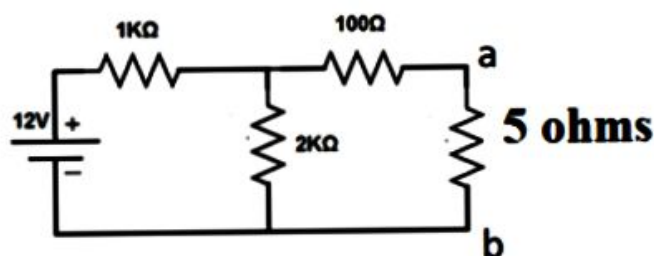


Figure 1

- b) Find the current 'i' in the circuit below figure 2.

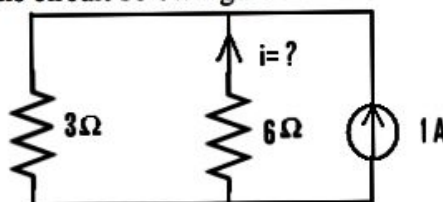
[6+4]

Figure 2

OR

- 3.a) State and explain Superposition Theorem.
b) Define time constant. Explain the time domain analysis of first order Series RL circuit. [5+5]

- 4.a) Find the impedance of series R-L-C circuit with $R=50\Omega$, $X_L=25\Omega$ and $X_C=10\Omega$.
b) Derive the average value, peak value, form factor for a sine waveform. [4+6]

OR

- 5.a) Derive an expression for average power in a single-phase circuit contains L element across sinusoidal voltage.
b) Give the relationship between phase voltage and line Voltage, phase current and line current for balanced three phase delta connected system. [5+5]

- 6.a) What are the various connections of three phase transformer?
b) Define voltage regulation, what is the meaning of zero regulation in transformer? [6+4]

OR

- 7.a) Explain types of losses that takes place in a transformer.
b) Draw and explain the equivalent circuit of a transformer. [5+5]

- 8.a) Explain the constructional details of three phase induction motor.
b) Draw torque slip characteristics of three phase induction motor. [5+5]

OR

- 9.a) Explain the constructional details of synchronous generator.
b) What are different methods to control speed of induction motor? [5+5]

- 10.a) Explain the types of batteries and its important characteristics.
b) Mention the importance of power factor improvement. [5+5]

OR

- 11.a) Write a short notes on battery back-up.
b) Explain how an MCB Works. [5+5]

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Code No: 151AH

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**B. Tech I Year I Semester Examinations, March/April - 2023****ENGLISH****(Common to EEE, CSE, IT, CSIT, CE(SE), CSE(CS), CSE(DS), CSE(N))****Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) Fill in the blank with the correct article [2]
i) Let us watch ___ sunset from that building.
ii) She doesn't even know how to use ___ computer.
- b) Fill in the blanks with the correct preposition. [3]
i) The degree is designed ___ build student's capacity.
ii) They need to concentrate ___ their studies to get through the exams.
iii) Current issues in the field should be addressed ___ means of a review of literature.
- c) Circle the synonyms in each sentence. [2]
i) The popular twins in school are fabulously liked by many.
ii) As we walked to school, the cool wind blew in our and is very chilly.
- d) Fill in the blanks with the words opposite in meaning to those underlined. [3]
i) Gold is an expensive metal while iron is
ii) My application was accepted but his was
iii) The teacher tried to make the student confident but he still looked very
- e) Rewrite the following sentences, making necessary corrections with regard to misplaced modifiers: [2]
i) Tarun almost found fifty cents under the sofa cushions.
ii) They bought a pet for my brother they call Puppy
- f) Go through the sentences and change their tenses as directed. [3]
i) The boy speaks the truth. (Present Continuous Tense)
ii) Mr. Crown has spoken about Dinosaurs. (Present Perfect Continuous Tense)
iii) The boat sailed yesterday. (Simple Present Tense)
- g) Write any two examples of redundant expressions. [2]
- h) Find out what the following abbreviations stand for. [3]
i) RSVP
ii) scuba
iii) TNT
- i) Write the meanings of the following technical terms. [2]
i) Firewall
ii) Buoyancy
- j) Who is the world's richest self-made woman? What was her childhood like? [3]

PART - B

(50 Marks)

- 2.a) What are the techniques of writing any composition precisely?
b) According to Raman, why is the sea in blue color? [5+5]
OR
- 3.a) Write a paragraph on *online education* in about 150 words.
b) Form adjectives by adding appropriate suffixes to the following words: [5+5]
person, create, beauty, excite, pain
- 4.a) What are rock-cut temples? Name some famous rock-cut temples in India.
b) Write a letter of complaint about the theft of your mobile phone. [5+5]
OR
- 5.a) Correct the errors in the following sentences with regard to concord.
i) She become very nervous whenever she see a river.
ii) Amrita uses a wheelchairs to move from one place to another.
iii) Neither you nor your brother are a singer.
iv) There are a box of chocolates.
v) Santhosh and his sister is going to the same school.
b) Write a letter of requisition to the Manager, *Toshiba Training Center*, asking about the courses they offer during summer vacation. [5+5]
- 6.a) How did the evolution of Blue Jeans take place?
b) Write the meanings of the following words borrowed from Latin language.
i) *bona fide*
ii) *quid pro quo*
iii) *de facto*
iv) *vice versa*
v) *ad hoc* [5+5]
OR
- 7.a) Describe the historical monument that you like the most.
b) Write the meanings of the following words borrowed from French language.
i) *chauffeur*
ii) *bon appetit*
iii) *faux pax*
iv) *en masse*
v) *restaurant* [5+5]

8.a) Write an essay on *Globalization and Its Impact* in about 200 words.

b) Make a precis of the following passage and give a suitable title.

Religion seems to me to be based primarily and mainly on fear. It is partly the terror of the unknown and the desire to feel that you have a kind of big brother who will stand by you in all your troubles and conflicts. Fear is the basis of the whole thing – fear of the mysterious, fear of failure, fear of death. Fear is the parent of cruelty and therefore, it is not surprising that cruelty and religion go together. For fear in this world is the foundation of these two things, that we now understand certain things, and can master them with the help of science, which has forced its way step by step against Christianity: against the churches, and against all the opposition of the old doctrines. Science can help us overcome this terrible fear that mankind has lived in for generations. Science can no longer teach us to look for allies in the sky but to look below at our own efforts to make this earth a habitable place. [5+5]

OR

9.a) What is the difference between abbreviations and acronyms? Give suitable examples.

b) What happens in your body when you consume sugary drinks? [5+5]

10. Rewrite the following sentences after making necessary corrections, if any.

a) Myself I am Suresh.

b) I am having four brothers and three sisters.

c) He do not have a laptop.

d) Does she has a car?

e) "Today office is there?"

f) That only, she is very arrogant.

g) I cannot cope up with this pressure.

h) I came to office by walk.

i) Our classroom is in the 2nd floor.

j) The price of this mobile phone is higher than yours. [10]

OR

11.a) What are the factors that made Ms Zhou successful?

b) Write a report on the fire accident occurred in your community. [5+5]

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Code No: 151AA

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech I Year I Semester Examinations, March/April - 2023

MATHEMATICS - I

(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, MMT, ECM, AE, MIE, PTM, CDS, CSIT, ITE, CE(SE), CSE(CS), CSE(AI&ML), CSE(DS), CSE(IOT), CSE(N), TTE, AI&DS, AI&ML, CSD)

Time: 3 Hours

Max. Marks: 75

Note: i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

- 1.a) Find the rank of the matrix by reducing to Echelon form $A = \begin{bmatrix} 1 & 4 & -2 & 1 \\ -2 & -3 & 4 & 3 \\ -3 & 3 & 6 & 12 \end{bmatrix}$ [2]
- b) Check whether the matrix $A = \begin{bmatrix} i & 0 & 0 \\ 0 & 0 & i \\ 0 & i & 0 \end{bmatrix}$ is Skew-Hermitian or not. [3]
- c) If $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$, then find the eigenvalues of $3A^3 + 4A^2 - 6A + 2I$. [2]
- d) If $A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$, then find A^3 using Cayley Hamilton theorem. [3]
- e) Test for convergence of the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n-1)}$. [2]
- f) Test for convergence of the series $\sum_{n=1}^{\infty} \log\left(1 + \frac{1}{n}\right)$. [2]
- g) Evaluate $\int_0^{\infty} x^2 e^{-x^2} dx$ using Beta-Gamma function. [2]
- h) Find the value of 'c' using Cauchy's mean value theorem for the function $f(x) = x^2$ and $g(x) = x^3$ [1, 2]. [3]
- i) If $x^3 + y^3 - 3axy = 0$ then find the value of $\frac{dy}{dx}$. [2]
- j) Using Euler's theorem find the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$, if $u = \frac{x^3 y^3}{x^3 + y^3}$. [3]

PART - B**(50 Marks)**

2.a) Find rank of matrix $A = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -2 & 1 \\ 1 & -1 & 4 & 0 \\ 2 & 2 & 8 & 0 \end{bmatrix}$ by reducing it to normal form.

b) Solve the system of equations $x - y + 2z = 4$, $3x + y + 4z = 6$, $x + y + z = 1$ by using Gauss elimination method. [5+5]

OR

3.a) Find the inverse of the matrix by using Gauss-Jordan method $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$.

b) Solve the system of equations $x + y + 54z = 110$; $27x + 6y - z = 85$; $6x + 15y + 2z = 72$ using Gauss Seidel method. [5+5]

4.a) Find the Eigen values and Eigen vectors of $A = \begin{bmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{bmatrix}$.

b) Verify Cayley Hamilton theorem for the Matrix $A = \begin{bmatrix} 1 & 3 & 2 \\ 1 & 9 & 1 \\ 3 & 7 & 0 \end{bmatrix}$ and hence find A^{-1} .

[5+5]

OR

5. Reduce the quadratic form $3x_1^2 + 3x_2^2 + 3x_3^2 + 2x_1x_2 + 2x_1x_3 - 2x_2x_3$ to canonical form using orthogonal transformation. Also find signature and rank of the quadratic form. [10]

6.a) Examine the convergence of $\sum \left[\frac{1.4.7 \dots (3n-2)}{3.6.9 \dots 3n} \right]^2$.

b) Examine the convergence of $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, (x > 0)$. [5+5]

OR

7.a) Test for convergence of $1 + \frac{3}{7}x + \frac{3.6}{7.10}x^2 + \frac{3.6.9}{7.10.13}x^3 + \dots$

b) Test for convergence of $\sum \left(\frac{n-1}{n} \right)^{n^2}$. [5+5]

8.a) If $f(x) = \sqrt{x}$ and $g(x) = \frac{1}{\sqrt{x}}$ prove that 'c' of the Cauchy's generalized mean value theorem is the geometric mean of 'a' and 'b' for $a > 0, b > 0$.

b) Find Maclaurin's series expansion of the $f(x, y) = \sin^2 x$. [5+5]

OR

9.a) Find the area bounded by pair of curve $y = 2 - x$ and $y^2 = 2(2 - x)$.

b) Evaluate $\int_{-\infty}^{\infty} e^{-a^2 x^2} dx$. [5+5]

10.a) Find $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ using Euler's theorem for the function $u = \cos^{-1} \left(\frac{x+y}{\sqrt{x} + \sqrt{y}} \right)$.

b) If $u = \frac{yz}{x}$, $v = \frac{xz}{y}$, $w = \frac{xy}{z}$ find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$. [5+5]

OR

11.a) Determine whether the functions $U = \frac{x}{y-z}$, $V = \frac{y}{z-x}$, $W = \frac{z}{x-y}$ are dependent. If dependent find the relationship between them.

b) Find the temperature at any point (x, y, z) in space is $f = 400xyz^2$. Find the highest temperature on the surface of the unit sphere $x^2 + y^2 + z^2 = 1$. [5+5]

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R18

Code No: 151AF

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**B. Tech I Year I Semester Examinations, March/April - 2023****CHEMISTRY****(Common to EEE, CSE, IT, CSIT, ITE, CE(SE), CSE(CS), CSE(DS), CSE(N), CSD)****Time: 3 Hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A**(25 Marks)**

- 1.a) Define the term ligand. Give examples. [2]
- b) Calculate the bond order N_2 . [3]
- c) What is calgon conditioning? Write the reaction involved. [2]
- d) How many grams of $FeSO_4$ dissolved per litre gives 210.5 ppm of hardness. [3]
- e) Give two factors affecting the rate of corrosion. [2]
- f) What is electro less plating? Explain briefly. [3]
- g) How is aspirin synthesized? Write its applications. [2]
- h) What is optical activity? Write the number of Enantiomers possible for Lactic Acid. [3]
- i) What is the role of TMS(Tetramethylsilane) in NMR spectroscopy? [2]
- j) Methane does not absorb IR radiations. Why? [3]

PART - B**(50 Marks)**

- 2.a) Explain the crystal field splitting of d -orbital's in Square planar complexes. [5+5]
 - b) Draw and explain the π - molecular orbital's of Benzene molecule. [5+5]
- OR**
- 3.a) What is Crystal Field Theory? Write the salient features of CFT. [5+5]
 - b) Discuss the crystal field splitting of d -orbital's in tetrahedral complexes. [5+5]
- 4.a) Write the causes and effects and preventive methods for caustic Embrittlement in boiler feed water. [5+5]
 - b) Write the principle involved in Reverse osmosis? Explain the desalination of Brackish water by Reverse Osmosis method. [5+5]
- OR**
- 5.a) What are the specification of Potable water? Write two methods of disinfection of Drinking water. [5+5]
 - b) A water sample on analysis gave the following data.
 $CaSO_4 = 60 \text{ mg/l}$, $Mg(HCO_3)_2 = 74 \text{ mg/l}$, $CaCl_2 = 24 \text{ mg/l}$, $NaCl = 5 \text{ mg/l}$, $MgCl_2 = 20 \text{ mg/l}$. Calculate Temporary, Permanent and Total hardness in degree Clark units. [5+5]

- 6.a) Explain with a neat diagram the construction and working of Calomel electrode.
b) What is cathodic protection? Discuss the sacrificial anodic method of protection of metals. [5+5]

OR

- 7.a) Describe the construction and working of Lithium ion battery.
b) Write the mechanism of electrochemical corrosion by taking the example rusting of iron. [5+5]
- 8.a) Explain the terms : Geometrical isomerism and Diastereomers with examples.
b) Discuss the mechanism of Markownikoff and anti Markownikoff's addition reaction of HBr to propene. [5+5]

OR

- 9.a) What is Grignard reagent? Discuss the mechanism of addition of Grignard reagent to on carbonyl compounds.
b) Discuss the reduction of carbonyl compounds by using LiAlH_4 and NaBH_4 reagents. [5+5]
- 10.a) Write the principle of UV-Visible spectroscopy. What type of compounds absorb UV- radiations? Give examples.
b) Write the stretching frequencies of the following functional groups in IR spectroscopy.
i) R-CHO ii) $\text{R-CH}_2\text{NH}_2$ iii) R-OH [5+5]
- OR
- 11.a) How do you distinguish the following compounds, cis-stilbene and Trans stilbene Compounds by UV spectroscopy:
b) Write the principle of ^1H NMR spectroscopy. Write the important application of NMR spectroscopy. [5+5]

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